Operating Systems: Process Management

A process is a program in execution. The Process Control Block (PCB) contains important information about the process including Process State, Program Counter, CPU registers, CPU scheduling information, and Memory-management information.

Process states include New, Ready, Running, Waiting, and Terminated. The process transitions from Ready to Running when selected by the short-term CPU scheduler.

CPU Scheduling Algorithms

CPU scheduling deals with the problem of choosing which process in the ready queue should be allocated the CPU core.

1. First-Come, First-Served (FCFS): Simple non-preemptive algorithm where the process requesting the CPU first is allocated first. Suffers from the Convoy Effect.
2. Shortest Job First (SJF): Associates each process with the length of its next CPU burst. It is provably optimal for minimizing average waiting time.
3. Round Robin (RR): Designed specifically for time-sharing systems. A small unit of time, called a time quantum (or time slice), typically 10 to 100 milliseconds, is defined. The ready queue is treated as a circular queue.
4. Priority Scheduling: A priority is associated with each process, and the CPU is allocated to the process with the highest priority. Major problem is Starvation (indefinite blocking), solved by Aging.

Deadlocks and Prevention

A deadlock is a situation where every process in a set is waiting for an event that can only be caused by another process in the set.

The Four Coffman Conditions for Deadlock:

1. Mutual Exclusion: At least one resource must be held in a non-shareable mode.
2. Hold and Wait: A process must be holding at least one resource and waiting to acquire additional resources held by other processes.
3. No Preemption: Resources cannot be preempted; a resource can only be released voluntarily by the process holding it.
4. Circular Wait: A closed chain of processes exists such that each process holds at least one resource needed by the next process in the chain.

The Banker\'s Algorithm is a deadlock avoidance algorithm for multiple instances of resource types.

Memory Management & Virtual Memory

Paging is a memory management scheme that eliminates the need for contiguous allocation of physical memory.

Physical memory is divided into fixed-sized blocks called Frames.

Logical memory is divided into blocks of the same size called Pages.

The Page Table maps logical page numbers to physical frame numbers.

Virtual memory allows execution of processes that are not completely in memory. It uses Demand Paging where pages are brought into memory only when they are needed during execution. A Page Fault occurs when a referenced page is not currently present in physical RAM.